

Designing an Architecture Diagram for a Storage Solution

In this exercise, we'll design an architecture diagram for an end-to-end data engineering storage solution. We'll consider the integration of services, the flow of data, and the use of appropriate storage technologies.

Learning Objectives

By the end of this train, you should be able to:

- Design an architecture diagram for an end-to-end data engineering storage solution.
- Map the integration of services involved in data ingestion, storage, and access.
- Identify appropriate storage technologies for different stages of the data flow.

Instructions

1. Open a whiteboard tool of your choice (Miro, Whimsical, Lucidchart, draw.io, etc.)
2. Use the prompt below to design your architecture diagram.
3. Label each component and annotate your diagram to explain data movement, storage, and technology choices.
4. Export your diagram as a PDF or image to upload if requested.

Your Scenario

You're working for a global logistics company that wants to modernise their data platform to support better analytics and faster decision-making. Your job is to design an architecture diagram that:

- Captures real-time package tracking data from IoT devices on trucks.
- Ingests batch delivery performance reports submitted daily from regional warehouses.
- Stores all raw data efficiently for compliance and archiving.
- Provides clean, transformed data to a BI team for reporting and dashboards.
- Supports data scientists who want to experiment with predictive models using historical data.

Your architecture must include:

- At least two data ingestion layers (one real-time, one batch).
- Raw and transformed data storage (consider options like S3, Delta Lake, Data Warehouse).
- ETL/ELT or streaming pipelines to clean and process the data.
- Integration points with reporting and data science tools (e.g. Power BI, Databricks, SageMaker).

Tips

- Use icons or shapes to distinguish between storage, compute, and processing layers.
- Clearly indicate the direction of data flow with arrows.
- Keep things modular — group components logically (e.g. "Ingestion", "Storage", "Processing").
- Make technology choices explicit — write the names (e.g. Kafka, Airflow, S3, Redshift, Spark, etc.)

Check Your Solution

Suggested Architecture:

- **Real-time ingestion:** Kafka or Kinesis for streaming IoT data
- **Batch ingestion:** Scheduled Airflow jobs or AWS Data Pipeline
- **Raw data storage:** Amazon S3, HDFS or Azure Data Lake (for compliance/archive)
- **Processing layer:** Spark jobs (for ETL), Delta Lake or Snowflake (for ELT)
- **Transformed data:** Stored in a Data Warehouse like Redshift, BigQuery or Snowflake
- **BI layer:** Connected tools like Power BI, Tableau, or Looker
- **Data science access:** Databricks, SageMaker, or Jupyter notebooks integrated via shared storage

Conclusion

Designing architecture diagrams is a vital part of data engineering. It helps visualize system complexity, align stakeholders, and ensure technical scalability. Mastering this skill empowers you to plan and build robust, efficient data infrastructure.

- **Takeaway 1:** Architecture diagrams guide technical decision-making and stakeholder communication.
- **Takeaway 2:** Choosing the right tools for ingestion, storage, and processing is critical to system success.
- **Takeaway 3:** Clear, modular diagrams help align business goals with data engineering solutions.